



A Counterfactual-History Engine for RWA Risk

BUIDL CTC 2026 Fall | RWA Track | Attestcoin Protocol

THE PROBLEM

On-chain RWA is transparent about ownership, but opaque about risk.

Tokenized real-world assets promise trustless transparency, yet the critical decisions - valuation, loan-to-value, default and liquidation - still come from a centralized appraiser or a black-box ML model. That trusted third party is exactly the failure mode tokenization was meant to remove.

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THE INSIGHT

A biological connectome is now a forkable, reproducible computational object.

MaleCNS v1.0 exposes the complete adult male fruit-fly central nervous system as machine-readable data: about 166,700 neurons, 25.6 million directed connections and 124 million synaptic contacts. Instead of trusting a rating agency, anyone can load the same graph, apply the same dynamics and re-run the same experiment.

WHAT COUNTERFLY DOES

It does not predict the future. It replays verified history and alternative futures.

Counterfly turns the connectome into a deterministic stress-testing primitive for RWA. It answers: what would this biological decision network have done under a rate shock, a missed payment, or a hazard? The answer becomes a reproducible risk action, not a proprietary score.

Attest	Attestcoin ProofBuilder + BlockProver verify a Sepolia RWA event on CC3.
Replay	The verified history and one counterfactual scenario replay through the connectome.
Commit	The action is EIP-712 signed and committed to the Counterfly ASC on CC3.
Write-back	A relayer bridge issues <code>adjustLtv</code> / <code>requestLiquidation</code> to a Sepolia action contract.

Three things make it memorable.

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Read, replay and ASC commit use the native SDK; write-back is an explicit relayer bridge.

The core decision loop is native: ProofBuilder and BlockProver attest the source event, the ASC stores the signed decision on CC3, and the relayer issues the cross-chain instruction. Native Attestcoin writability is still in third-party testing on CC3 testnet, so worker/src/writability.ts keeps the outbox/inbox interface ready while worker/src/relay.ts provides the auditable production fallback.

Read	ProofBuilder + BlockProver precompile
Compute	Deterministic MaleCNS replay, hash-pinned and open-source
ASC commit	CounterflyASC stores EIP-712 decision on CC3
Write-back	Relayer bridge -> Sepolia RwaAction

REPRODUCIBILITY

Same graph + same scenario + same seed = same decision.

Every replay is pinned by a graph hash and scenario hash. The demo ships a single-click recommended run: full MaleCNS v1.0 + RATE_SHOCK magnitude 0.8, which lands on LIQUIDATE. The worker, dashboard and contracts are all in one repository with CI.

Neurons	166,700
Directed edges	25,582,938
Synaptic contacts	124,177,617
Recommended demo	MaleCNS v1.0 / RATE_SHOCK 0.8 -> LIQUIDATE

DEPLOYMENT

Everything is live on testnet.

CounterflyASC v2 (CC3)	0x18bC3211e788d7fcb06B7aF968588Ee613C0b135
RwaAction (Sepolia)	0x4a1c9031ab8f736C4fEc8488b1294928EEE99817
Demo	https://counterfly.vercel.app
Fallback	http://144.91.75.120:8786

Built for CEIP-scale execution, not a one-off hackathon demo.

Counterfly ships as open infrastructure: reproducible replay, Attestcoin-native read path, and ASC-gated commits. Next milestones: native Attestcoin writability swap-in, additional source chains, issuer-facing APIs, and Creditcoin credit-rail integration for on-chain RWA underwriting history.

Now	CC3 ASC + Sepolia RwaAction + relayer write-back
Q4 2026	Native outbox/inbox when CC3 exports writability
2027	Issuer SDK + credit-history linkage on Creditcoin

Top-three teams receive CertiK audit credits; we designed for reviewability.

Reads are Attestcoin-verified; commits are EIP-712 worker-gated; scenarios require a registered source tx. Relay keys are separate from worker signing keys. Full MaleCNS replay is hash-pinned and covered by CI golden tests (see docs/REPRODUCE.md).

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WHY IT SCORES

A direct answer to the RWA trust gap, built on Attestcoin.

Counterfly maps the official scoring dimensions to a single coherent demo: a meaningful Attestcoin integration, a genuinely original RWA idea, a working testnet deployment, and a reproducible open-source pipeline. Record the 90-second demo using docs/DEMO_VIDEO.md.